

RITESH PANDEY

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🔗 [LeetCode](#) | [Codeforces](#) | [CodeChef](#) | [Portfolio](#)

ACHIEVEMENTS & HONORS

- **ICPC 2025 Regionalist** – Qualified prestigious onsite regionals at Amritapuri & Chennai – **Rank 81 at Chennai** and **Rank 284 at Amritapuri** among nationwide teams.
- **LeetCode**: Knight (Max Rating 2001) – Global Rank **589** in Biweekly Contest 168.
- **Codeforces**: Specialist (Max Rating - 1419) – Global Rank **1171** in Round 1054.
- **CodeChef**: 3-Star (Max Rating - 1703) – Global Rank **197** in Starters 177.
- Solved **1500+** Data Structures & Algorithms problems; participated in **100+** contests.

EDUCATION

Madan Mohan Malaviya University of Technology
B.Tech in Computer Science and Engineering (CGPA: 8.59 / 10.0)

Gorakhpur, U.P., India

Sep 2023 – May 2027

EXPERIENCE

ZoAstro Tech Pvt. Ltd. (HRDYAM)

June 2026 – Present

Full Stack Developer Intern

Remote

- Engineered a production-grade **RAG-based AI chatbot** integrating Gemini embeddings and Sanity CMS across 4 core domains, reducing unresolved user queries by **40%** via low-latency semantic vector retrieval.
- Architected an **AEO Knowledge Graph** featuring structured Q&A pages and JSON-LD FAQ schema markup spanning 200+ intent queries, securing citation visibility by major AI search engines (Perplexity, ChatGPT, Gemini).
- Built a high-throughput multilingual NLP pipeline (Hindi/English auto-detection) with 3-tier intent classification and automated WhatsApp handoff, cutting manual support intervention overhead by **20%**.

PROJECTS

SoulCare | [Express.js](#), [MongoDB](#), [Firebase](#), [RAG](#), [ChromaDB](#), [WebRTC](#) ([GitHub](#) · [Demo](#)) Nov 2025

- Designed and built a scalable mental health platform with Role-Based Access Control (RBAC) across 3 distinct user tiers, securing REST API endpoints via encrypted JWT middleware and custom route guards.
- Implemented an end-to-end **RAG pipeline** utilizing ChromaDB vector store and Gemini embeddings with 3-tier intent classification, delivering context-aware peer support responses across 500+ concurrent user queries.
- Engineered real-time anonymous peer communication channels via **WebSockets and WebRTC** alongside an automated clinical assessment module for real-time mood risk-level prediction and session scheduling.

Society HomeChef | [MERN](#), [Socket.io](#), [Gemini AI](#), [Concurrency Control](#) ([GitHub](#) · [Demo](#)) Mar 2026

- Architected a multi-role food delivery engine (chef, customer, rider) powered by a custom **Socket.io real-time dispatch system** handling high-frequency concurrent order streams seamlessly.
- Eliminated race conditions and concurrency anomalies using **atomic MongoDB transactions** across parallel order placement flows, guaranteeing absolute data consistency under 50+ simultaneous requests.
- Integrated cryptographic OTP verification for tamper-proof delivery handoffs and Gemini AI-driven nutritional analysis equipped with robust heuristic fallback layers for 99.9% uptime.

TECHNICAL SKILLS

- **Languages**: C++, Java, Python, JavaScript, TypeScript, SQL
- **Frameworks & Backend**: React.js, Next.js, Node.js, Express.js, REST APIs, WebSockets, WebRTC, Tailwind CSS
- **GenAI & AI Systems**: RAG, Vector Embeddings, Prompt Engineering, ChromaDB, Semantic Search, LLM Integration
- **Databases & Storage**: MongoDB, MySQL, Supabase, Firebase, Sanity CMS
- **Architecture & Concepts**: Distributed Systems, Concurrency, Database Transactions, RBAC, System Design
- **DevOps & Tools**: Git, GitHub, Docker, CI/CD, Linux, Postman, VS Code

RELEVANT COURSEWORK

- Data Structures & Algorithms, Object-Oriented Programming, Operatin Systems, Database Management Systems, Computer Networks, Distributed Systems, Software Engineering, Artificial Intelligence, Machine Learning, Linux.

LEADERSHIP & ADDITIONAL ACHIEVEMENTS

- **DSA Head, MMMUT Reso** – Mentored **200+** students in Data Structures & Algorithms and Competitive Programming, supporting peer learning and technical development.
- **Global Contest Rankings** – Meta Hacker Cup 2025 Round 2, **1598**; TCS CodeVita Season 13: **1670** among **100,000+** global participants.